



INSTAGRAM USER BEHAVIOUR AND INFLUENCER MAPPING

A Data Driven Study Using SQL

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AGENDA

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- Hashtag Popularity
- Low Engagement User Hashtag Trends
- Strategies to Engage Inactive Users



AGENDA

- Strategies to Engage Inactive Users
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ABOUT INSTAGRAM

Instagram is a global social media platform focused on photo, video, and short-form content sharing. Launched in 2010, it has evolved into a space where individuals, creators, and businesses showcase visual stories, engage with audiences, and build communities. With features like Stories, Reels, IGTV, and direct messaging, Instagram encourages real-time interaction and creative expression. Today, it serves as a powerful tool for digital marketing, brand building, and trend discovery, making it one of the most influential platforms in the social media ecosystem.

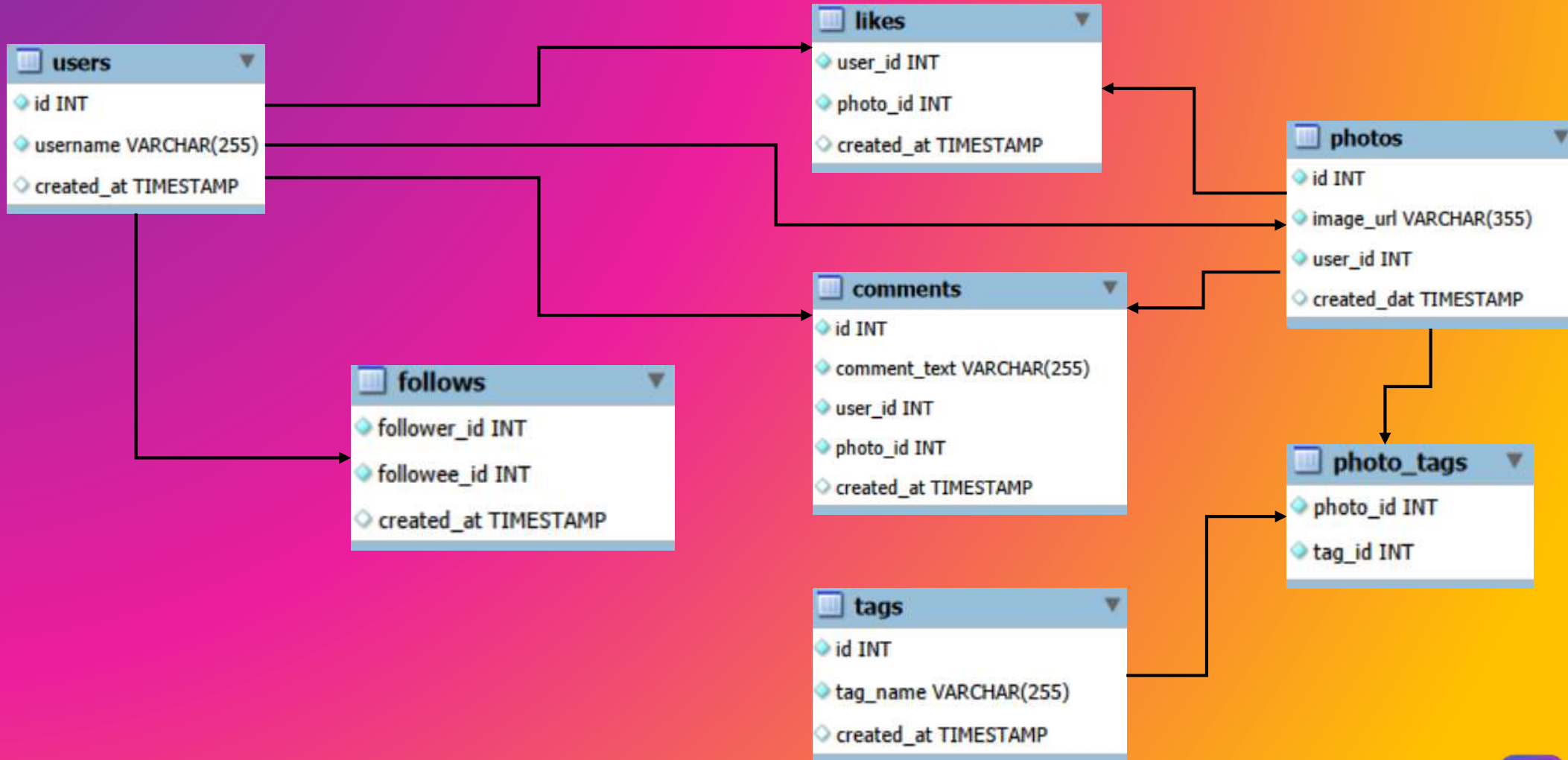


BUSINESS CHALLENGE

Instagram aims to strengthen its marketing impact by utilizing user-level insights more effectively. The objective is to analyze platform data to enhance targeted marketing strategies that drive higher user engagement, retention, and acquisition.

This can be achieved by examining historical user interactions, engagement patterns, and behavioral trends to identify key opportunities for optimization and support data-driven decision-making for sustained platform growth.

DATABASE SCHEMA



USER BEHAVIOUR OVERVIEW



Data Overview

Dataset Size: 80 user

Engagement Records: 1,500+ interactions processed for scoring

Segmentation Output: 4 influencer tiers identified - Top, Strong, Growing & Emerging

Average Engagement Score: Ranged from **3.25** to **11**

Follower Distribution: Majority of profiles clustered around **76–77** followers

Activity Depth: Total posts per user ranged from **1** to **12**

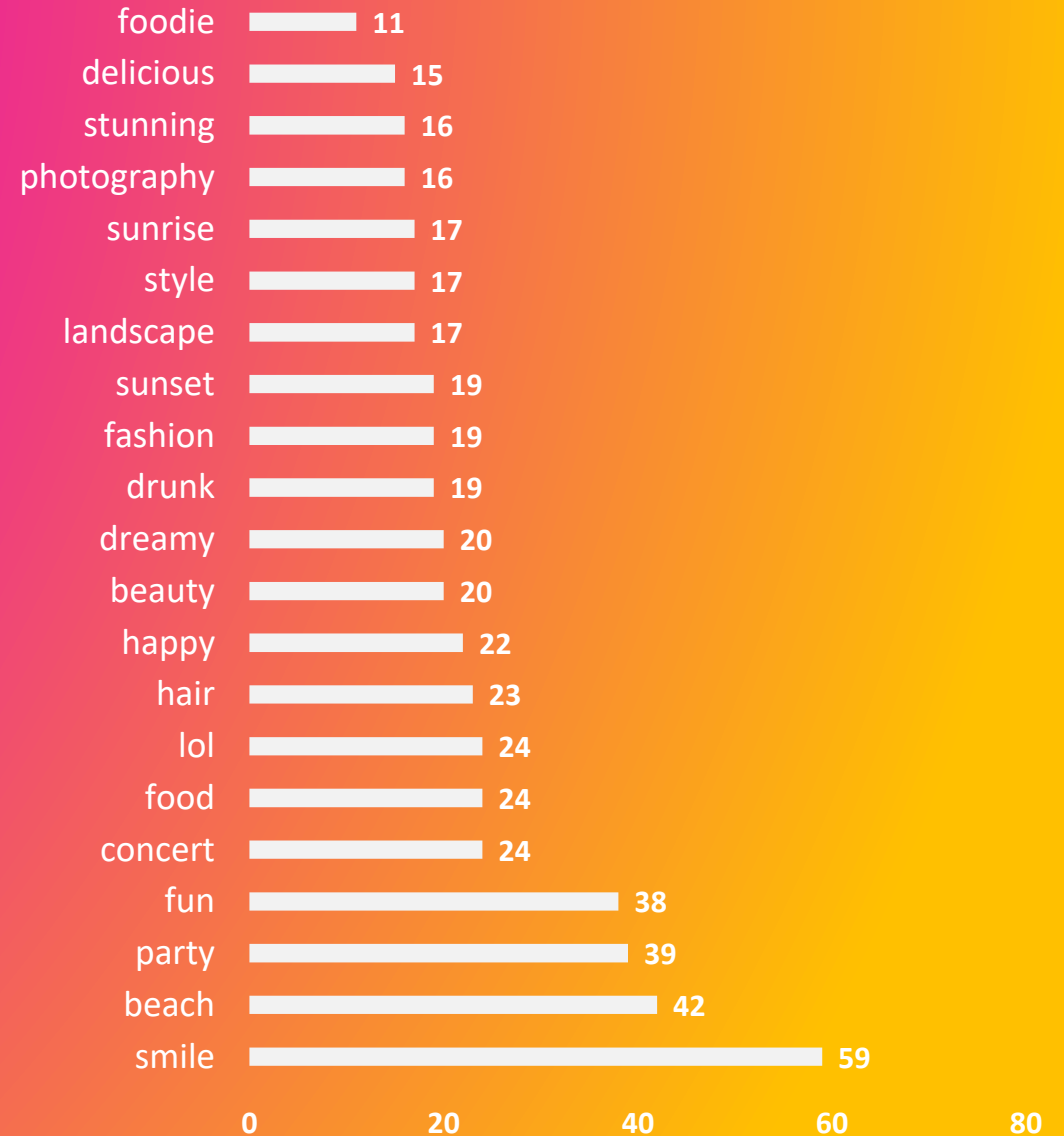
Key Insight: High-engagement creators are significantly outperforming others even with small follower bases, making them strong candidates for brand advocacy, micro-campaigns, and content beta testing.

DATA PREPROCESSING

- Duplicate records were detected and removed using **distinct table recreation**, ensuring unique row integrity across all tables.
- **Null value checks** were performed column-wise using SUM(IS NULL), confirming **no missing data** in the dataset.
- Referential links between users, photos, likes, comments, follows, photos, photo_tags and tags tables were validated for **schema consistency**.
- Analytical tables such as **user_activity**, **followers_list**, **user_and_tags**, **engagement_range**, and **tag effectiveness metrics** were created to support engagement and influencer insights.

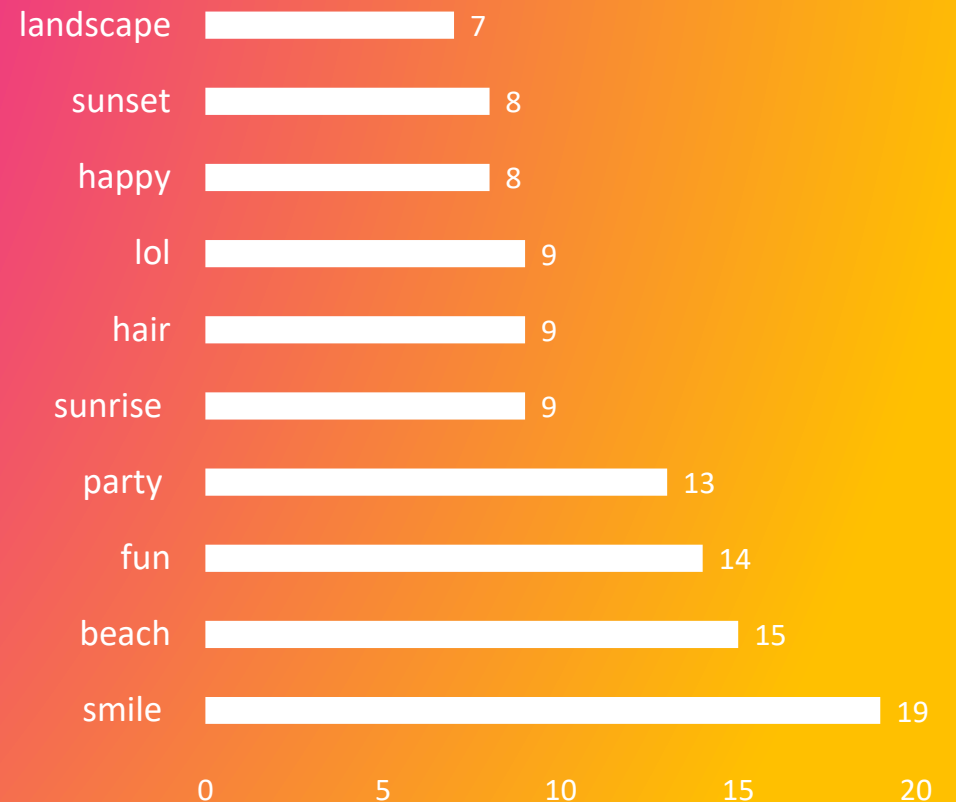
HASTAGS POPULARITY

- **#smile** is the most widely adopted tag with **59 users**, indicating strong preference for joyful and expressive content.
- Tags like **#beach**, **#party**, **#fun** show that users are highly engaged with **social and leisure-oriented themes**.
- **#concert**, **#food**, **#lol**, **#hair** follow closely, reflecting a mix of entertainment and lifestyle content trends.
- Emotional and aesthetic tags such as **#happy**, **#beauty**, **#dreamy** remain consistently used across the platform.
- Visual-centric tags **#photography**, **#sunset**, **#landscape** show continued interest in scenic and creative posts.
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LOW-ENGAGEMENT USER HASHTAG TRENDS

- Low-engagement users were identified as users **who post content but do not like or comment on posts.**
- Hashtag usage of this segment was analyzed to understand their content preferences.
- The top 10 most frequently used hashtags among low-engagement users were identified: *Smile (19)*, *Beach (15)*, *Fun (14)*, and *Party (13)* emerged as the most used hashtags.
- ***Smile* and *Party* are also trending among the general user population.**
- These hashtags can be leveraged to push relevant content and encourage higher engagement.



STRATEGIES TO ENGAGE INACTIVE USERS



Personalized Content Feed

Deliver content tailored to users' frequently used hashtags to ensure relevance and increase content visibility.



Creator-led Content

Promote high-quality posts from creators within relevant hashtag communities to leverage credibility, social proof, and inspire interaction among inactive users.



Contextual Notification

Trigger timely notifications when trending or newly published content appears within user's interest-based hashtags.

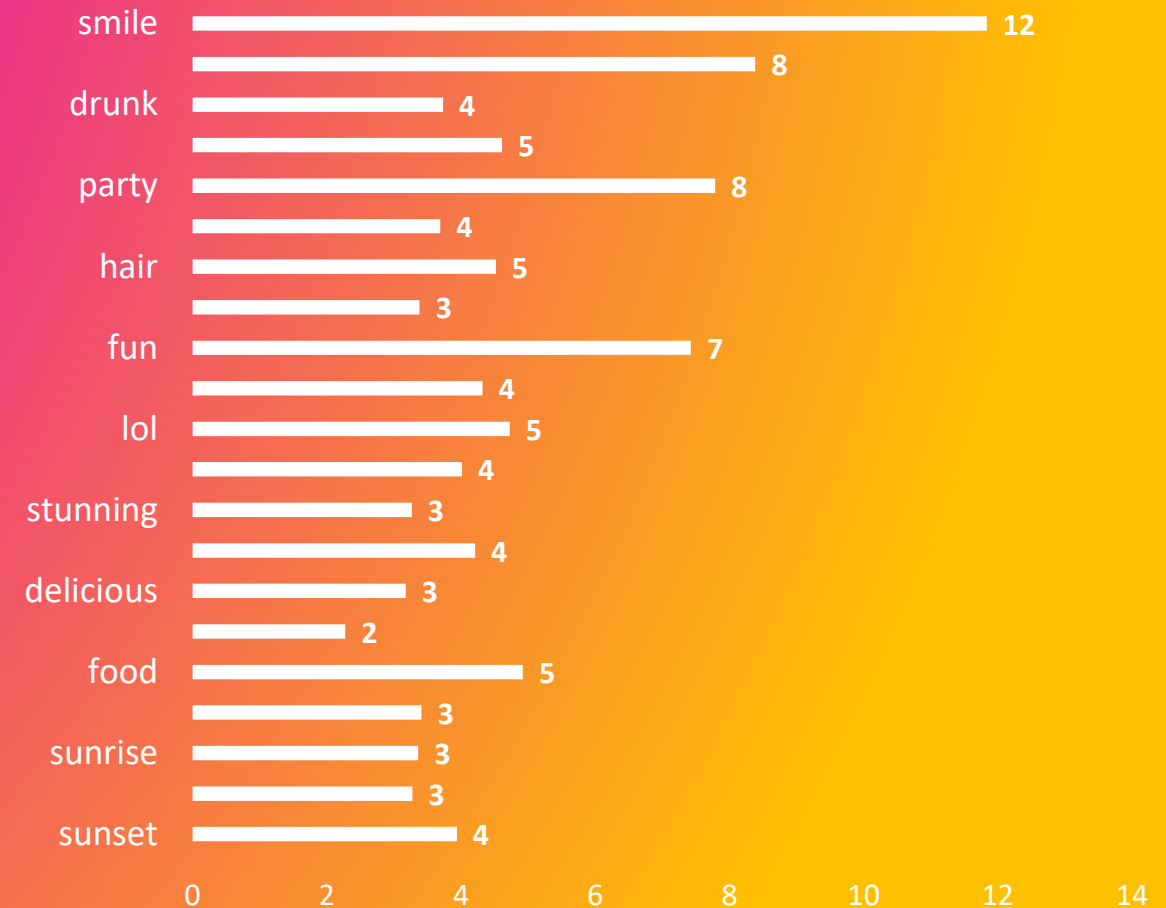


Community Clustering

Group users with similar hashtag usage patterns to surface peer content and foster a sense of community participation.

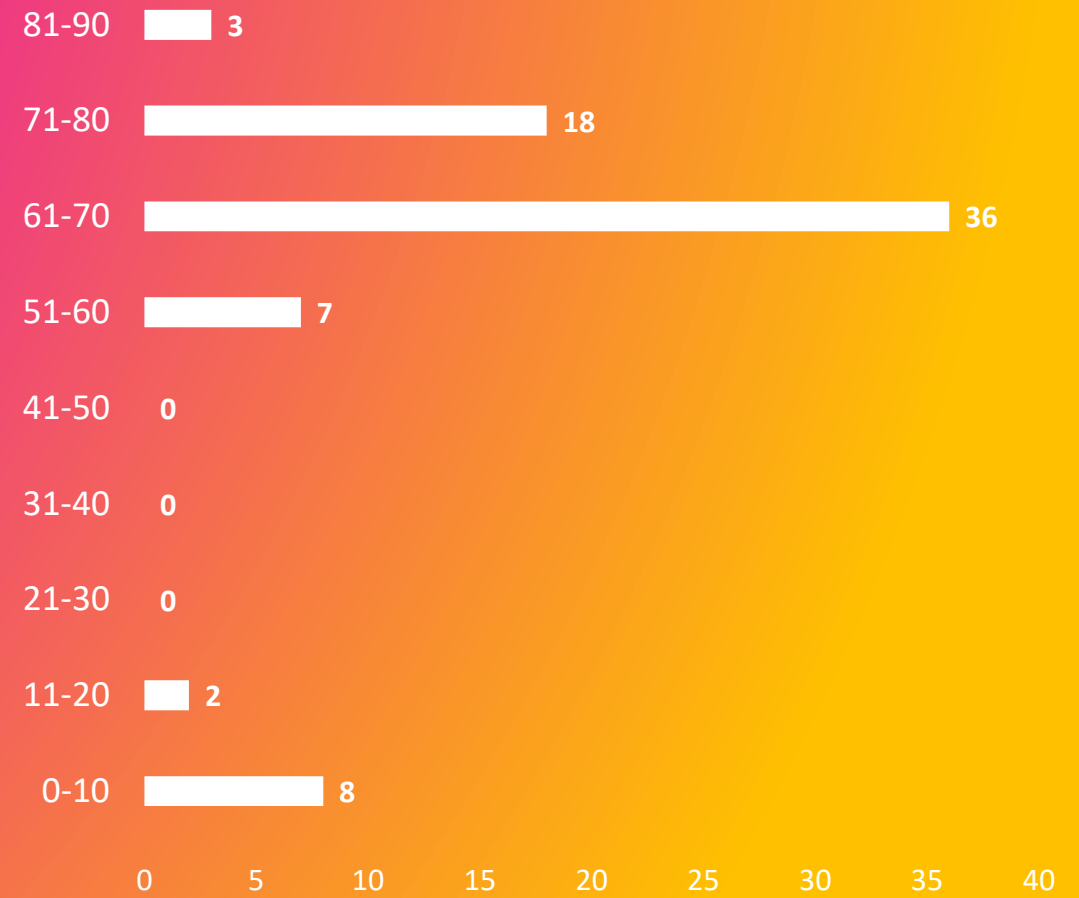
HASTAGS EFFECTIVENESS

- Effectiveness here is measured by combining likes and comments generated by each hashtag. (Scores were in large numeric values, so they were scaled down for cleaner visualization.)
- The hashtags **#smile**, **#party**, and **#beach** deliver the **strongest engagement**, making them ideal focal points for high-impact content.
- Tags related to **fun, food, beauty, and travel themes** (e.g., #fun, #sunset, #beauty, #fashion) also show strong performance, indicating audience preference toward lifestyle-driven content.
- Hashtags with moderate performance like **#foodie**, **#style**, **#photography** present opportunities for optimization through content enhancements.



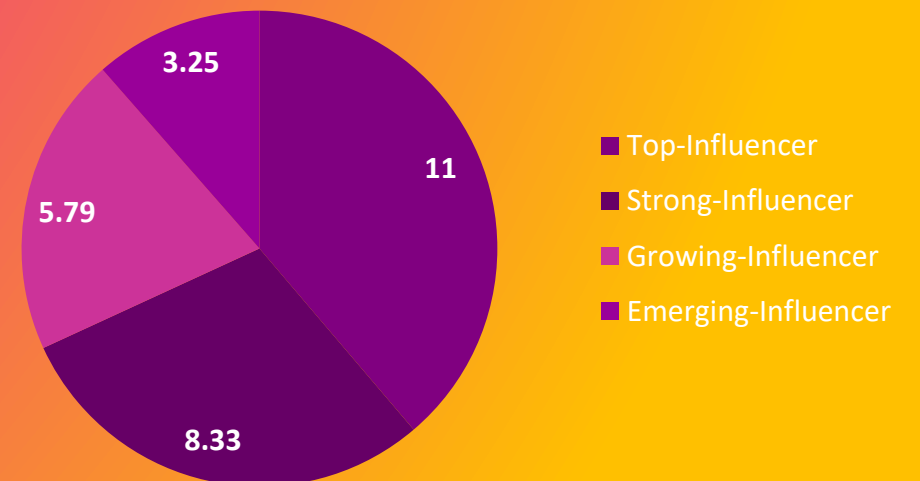
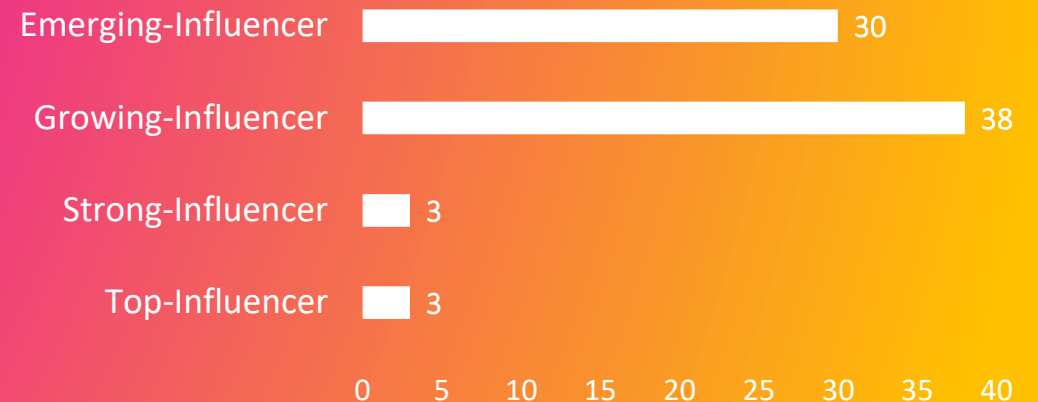
ENGAGEMENT SCORE DISTRIBUTION

- Engagement score is calculated as the sum of each user's **total posts, total likes, and total comments**.
- The majority of users fall within the **61 - 70 score range (36 profiles)**, indicating a strong presence of moderately active users on the platform.
- Another notable segment is **71 - 80 (18 profiles)**, showing higher activity levels among the next group.
- Very few users reach the **81 - 90** range (3 profiles), and only small numbers fall into the **0 - 10** and **11 - 20** ranges.
- The absence of users in the **21 - 50** ranges indicates a sharp jump in user engagement patterns.
- Overall, the platform demonstrates a **healthy mid-to-high engagement base** with strong potential for conversion into highly active users.



POTENTIAL INFLUENCE MAPPING

- Influencers are segmented (in a barchart) using **Engagement Score** → $(\text{Posts} + \text{Likes} + \text{Comments}) \div \text{Followers}$ to ensure only **authentic engagement** is measured.
- Majority of users fall under **Growing (38)** and **Emerging (30)** tiers, indicating a strong pool of rising creators.
- Only **3 Top** and **3 Strong** influencers exist, showing limited high - impact reach at present.
- The pie chart below highlights engagement quality differences across tiers: Top (11), Strong (8.33), Growing (5.79), and Emerging (3.25).
- Strategy: Prioritize **Top & Strong** influencers for campaigns while **nurturing mid-tier users** into future high-impact ambassadors.



CONCLUSION

- User engagement varies significantly, while **a few users drive high interaction**, many exhibit **passive behavior** (likes/comments remain low despite posting).
- **Growing & Emerging influencers dominate the platform** (68 out of total), showing a strong potential creator base still developing audience trust and reach.
- Engagement Score analysis confirms that **authentic influence is limited** to a very small segment (Top + Strong influencers = 6 users).
- A subset of users (10 profiles) show **zero engaging actions** despite initial posting, indicating **early churn behavior**.
- Content preferences are clear, **high-appeal hashtags** (#smile, #beach, #party, #fun) consistently outperform others, influencing user engagement patterns.

RECOMMENDATION & NEXT STEPS

- **Double-down on Top & Strong Influencers** for high-impact campaigns and brand-driven content amplification.
- **Develop Growing & Emerging Influencers** through structured programs, highlight them in explore feeds, feature community badges, and provide creator analytics support.
- **Re-engage passive users** via curated recommendations based on their **initial tag interests**, and trigger interactions with reminders or gamified rewards.
- **Push content aligned with top-engaging hashtags** to maintain high session time and lift interaction rates.
- Continuously **track engagement score evolution** to identify users transitioning upward in influence, these become **future brand advocates**.

THANK YOU

Open to Questions

